Beam Me Up: Practice Test 1

Time: 50 minutes (the challenge happens during this time). Work as a team of 3 and write one set of answers. You may bring only pencils and 1 protractor (no light sources, mirrors, timers, phones, or smartwatches). When the judge calls your team for the mirror challenge, stop, go do it, then come back and finish the test.

Partner names: _____ and _____ Score: _____

Part 1: Reflection and mirrors

Remember: angles are measured from the normal line, an imaginary line straight out from the mirror at 90 degrees.

1. Reflection is when light:

◯ A Soaks into a surface ◯ B Passes through a surface ◯ C Bounces off a surface
◯ D Bends as it enters water

2. A beam hits a mirror 25 degrees from the normal. What is the angle of reflection?

3. A beam hits a mirror 70 degrees from the mirror's surface. What is the angle of incidence (from the normal)? What is the angle of reflection?

4. The beam in and the beam out make a 100 degree angle with each other. What is the angle of incidence?

5. A beam comes in 40 degrees from the normal. What is the whole angle between the beam in and the beam out?

6. On the floor protractor, the beam reaches the wall mirror from 30 degrees on the RIGHT. Where will it go after it bounces?

7. You tilt your mirror just 5 degrees. About how far does the reflected beam turn?

◯ A 0 degrees ◯ B 5 degrees ◯ C 10 degrees (the beam turns twice as much as the mirror)
◯ D 45 degrees

8. A mirror and a sheet of white paper both reflect light. Why can you see yourself in the mirror but not in the paper?

9. The word AMBULANCE is written backwards on the front of an ambulance. Why?

Part 2: Refraction

10. Refraction is when light:

◯ A Bounces back off a mirror ◯ B Bends as it goes from one material into another
◯ C Turns into heat ◯ D Is blocked by an object

11. Why does light bend when it goes from air into water?

◯ A It changes speed (it slows down in water) ◯ B Water is magnetic ◯ C Water is dark
◯ D The light gets hot

12. White sunlight goes through a prism and comes out as a rainbow of colors. Which light word explains this?

13. A magnifying glass makes small things look bigger. What does its curved glass lens do to light?

14. A fish in a pond looks like it is in a different spot than it really is. Which light word explains this?

◯ A Reflection ◯ B Refraction ◯ C Absorption ◯ D Transmission

Part 3: Absorption, transmission, and scattering

15. Absorption is when light:

◯ A Bounces off ◯ B Passes straight through ◯ C Bends
◯ D Soaks into a material and turns into heat

16. On a sunny day, which color car will get hottest inside?

◯ A Black ◯ B White ◯ C Silver ◯ D They all get equally hot

17. A red apple looks red because it:

◯ A Makes its own red light ◯ B Reflects red light and absorbs the other colors ◯ C Absorbs red light
◯ D Lets red light pass through it

18. Transmission is when light:

◯ A Bounces off a material ◯ B Is soaked up by a material ◯ C Passes through a material
◯ D Stops moving

19. Label each as transparent, translucent, or opaque: (a) clear drinking glass, (b) frosted bathroom window, (c) wooden door.

20. Scattering is when light:

◯ A Bounces off tiny bits (dust, water drops, air) in many directions ◯ B Goes in one straight line
◯ C Turns into heat ◯ D Bends into a rainbow

21. Why does the sky look blue on a sunny day?

22. Sunsets often look red and orange. Why?

23. In the dark challenge room, you can see a flashlight beam in the air only if there is dust or mist. Why?

24. A green leaf in sunlight reflects green light and soaks up red and blue light. Name the two light words.

Part 4: The eye

25. Which part is the clear dome on the front of the eye that does most of the bending of light?

◯ A Sclera ◯ B Cornea ◯ C Retina ◯ D Iris

26. Which part is the tough white outer coat that protects the eye?

◯ A Lens ◯ B Pupil ◯ C Sclera ◯ D Conjunctiva

27. Pinkeye happens when the thin clear skin over the white of the eye and inside the eyelids gets red and swollen. What is that skin called?

◯ A Conjunctiva ◯ B Cornea ◯ C Optic nerve ◯ D Vitreous humor

28. In a dark room, what happens to your pupils, and which eye part does it?

29. The pupil is really:

◯ A A black dot painted on the eye ◯ B A hole in the middle of the iris that lets light in
◯ C The same thing as the lens ◯ D A muscle

30. You look up from your book to a tree far away. Which part pulls to change the shape of the lens so you can focus?

◯ A Iris ◯ B Optic nerve ◯ C Retina ◯ D Ciliary muscle

31. What is the job of the lens?

32. What is the clear jelly that fills the inside of the eye and keeps it round?

33. Where are the photoreceptors (rods and cones) found?

◯ A The cornea ◯ B The iris ◯ C The retina ◯ D The sclera

34. What is the job of the photoreceptors?

35. Cones or rods? (a) help you see colors, (b) help you see in dim light, (c) see fine detail, (d) see only black, white, and gray.

36. Each eye has a small blind spot with no rods or cones. It is where which part leaves the back of the eye?

◯ A Lens ◯ B Optic nerve ◯ C Ciliary muscle ◯ D Iris

Part 5: Putting it together

37. Light travels from a lamp, bounces off a book, and enters your eye. Name the path through the eye, in order, from the first part it enters to the part that carries the message to the brain.

38. Why can you see a book in a room, even though the book does not make its own light?

39. A periscope uses two mirrors to see over a wall. Which light word does it use?

40. Why is it hard to tell the color of a car at night, even under a dim streetlight?

31	It changes shape to focus light onto the retina.
32	Vitreous humor.
33	C The retina
34	They catch light and change it into signals that go to the brain.
35	(a) cones, (b) rods, (c) cones, (d) rods.
36	B Optic nerve
Part 5: Putting it together	
37	Cornea, pupil, lens, vitreous humor, retina (rods and cones), optic nerve.
38	Light from the lamp or sun reflects off the book into your eyes.
39	Reflection (each mirror bounces the light).
40	In dim light only the rods work well, and rods do not see color. The cones need bright light.